

Plan and conduct repeatable investigations to answer questions including, as appropriate, deciding the variables to be changed, measured and controlled in fair tests; describing potential risks; planning for the safe use of equipment and materials; and identifying required permissions to conduct investigations on Country/Place

Learning intention

- plan and conduct repeatable investigations to answer questions including, as appropriate, deciding the variables to be changed.

Teach and model

- determining which is the variable being tested and which variable is being measured.
- considering different ways to approach investigations including researching, using trial and error, experimental testing, field observations.
- identifying potential risks to themselves or others when conducting an investigation and explaining rules for safe processes and use of equipment and materials.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.